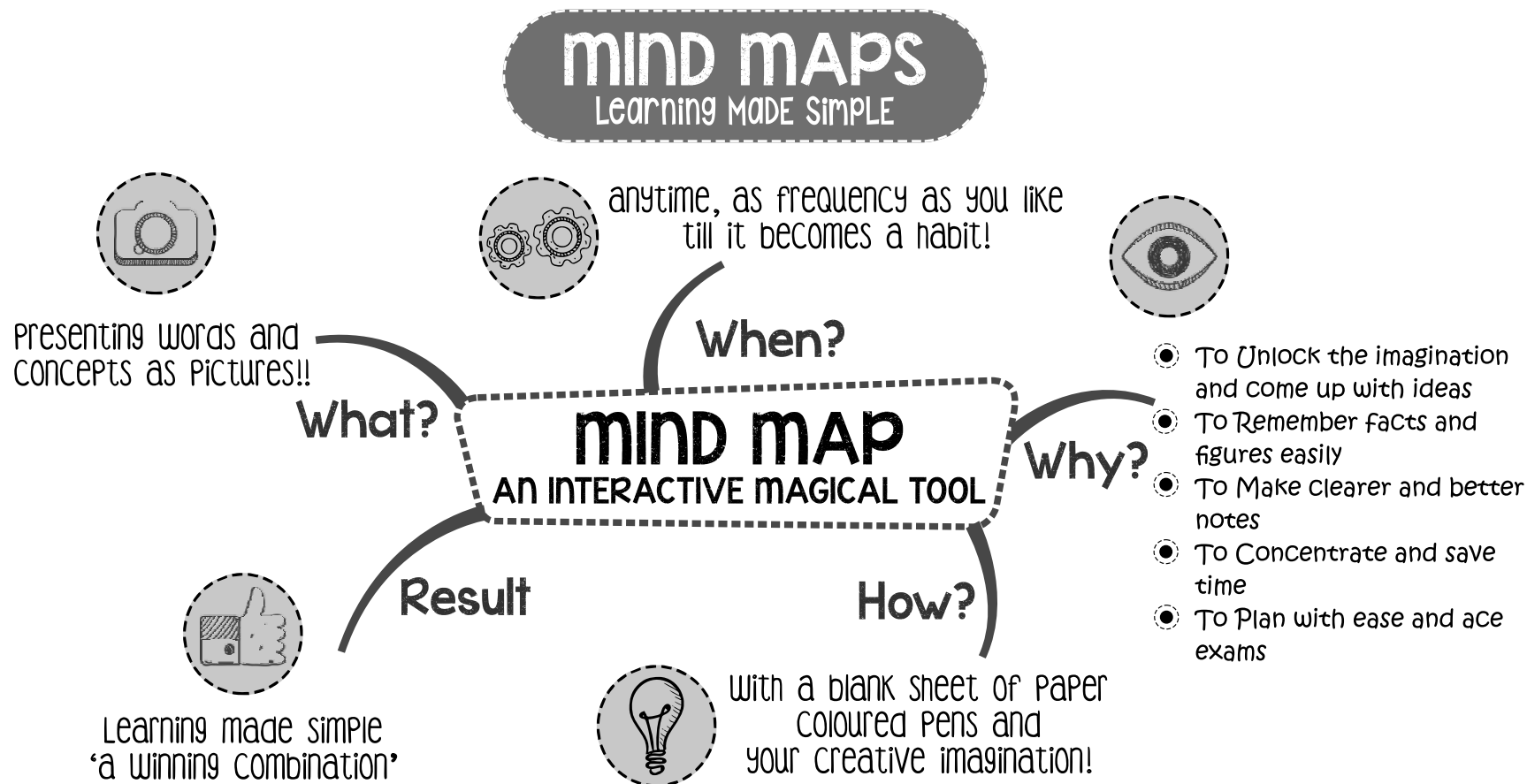
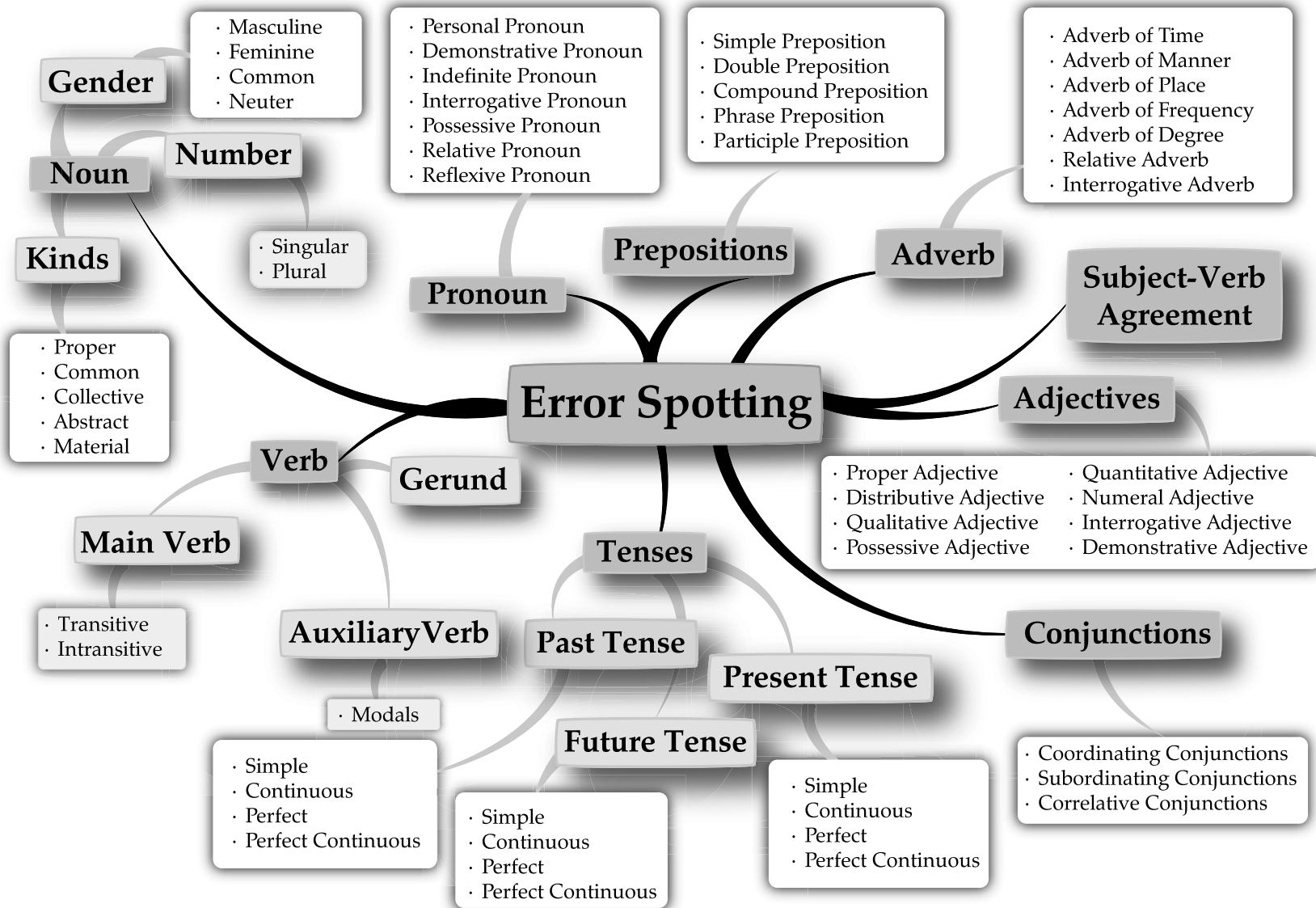


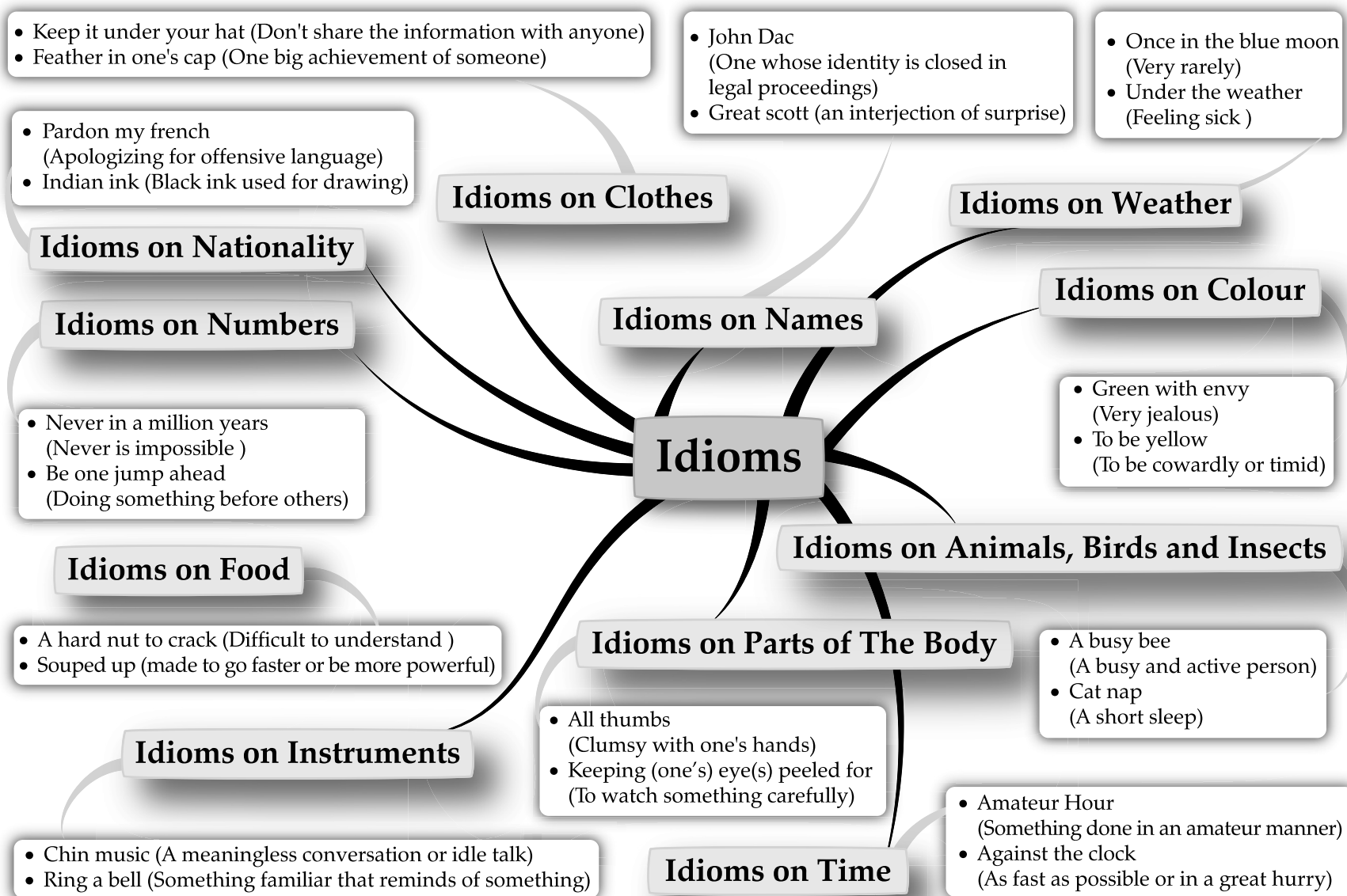


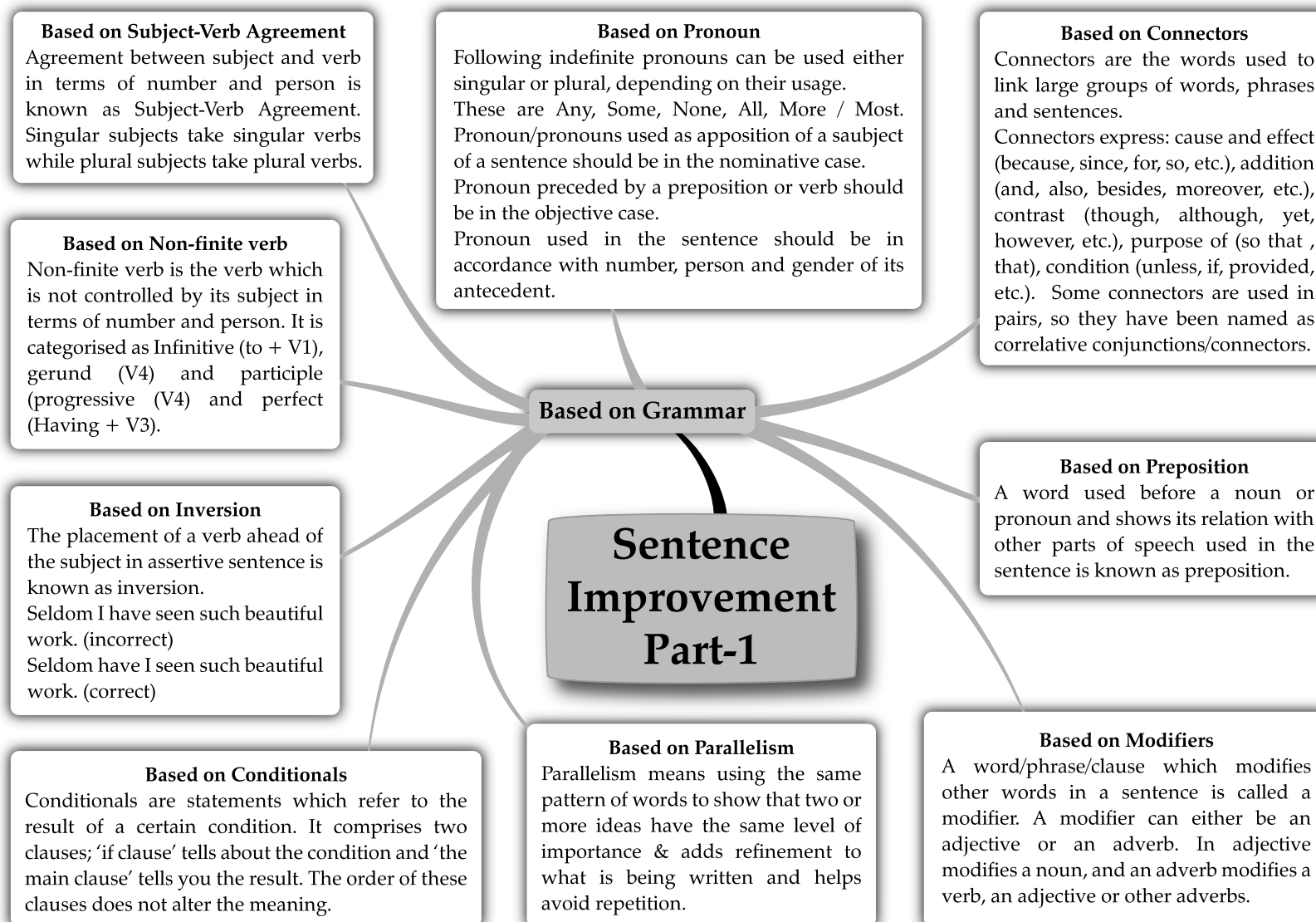
What are Associations?

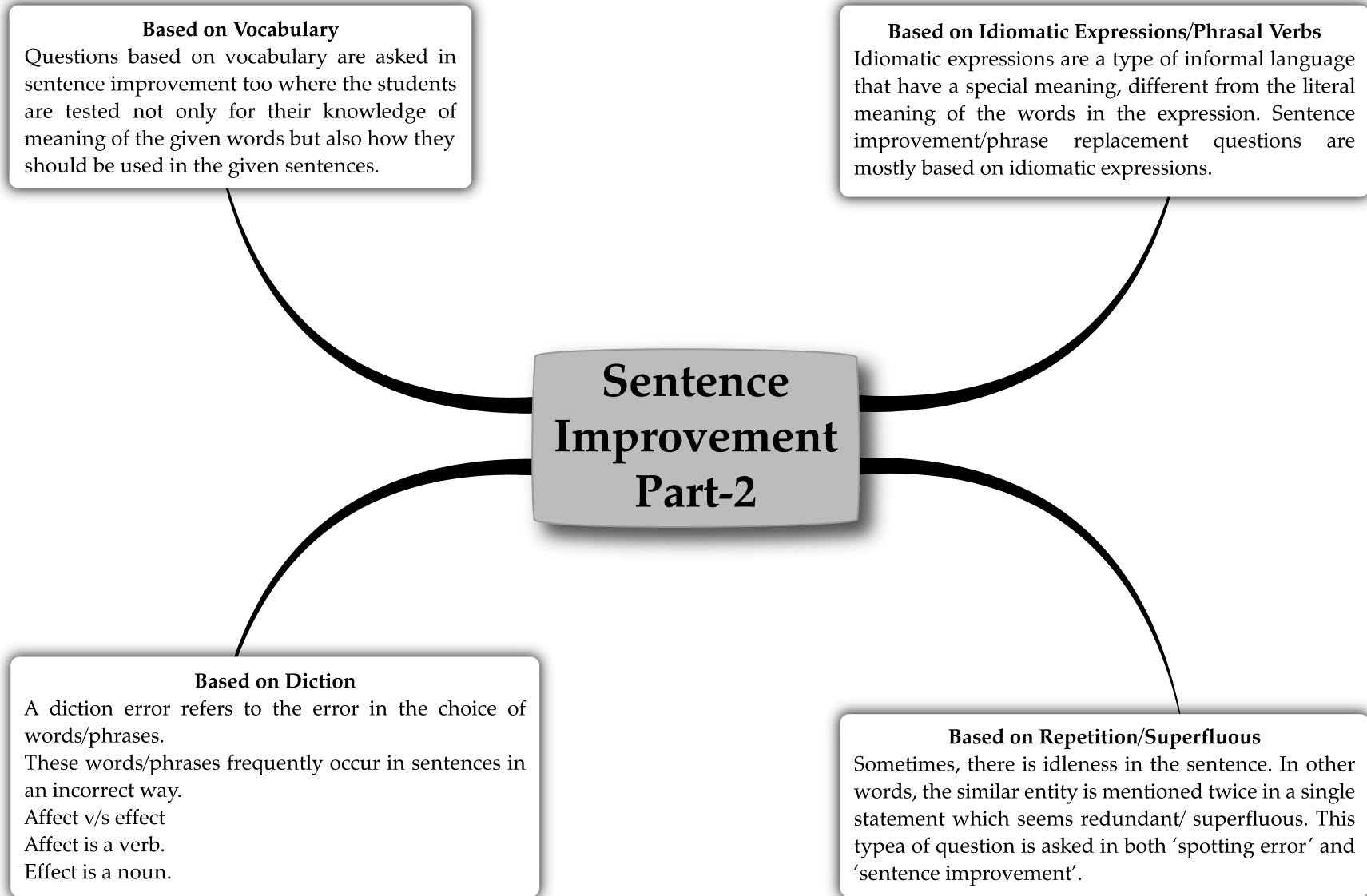
It's a technique connecting the core concept at the Centre to related concepts or ideas. Associations spreading out straight from the core concept are the First Level of Association. Then we have a Second Level of Association emitting from the first level and the chronology continues. The thickest line is the First Level of Association and the lines keep getting thinner as we move to the subsequent levels of association. This is exactly how the brain functions, therefore these Mind Maps. Associations are one powerful memory aid connecting seemingly unrelated concepts, hence strengthening memory.

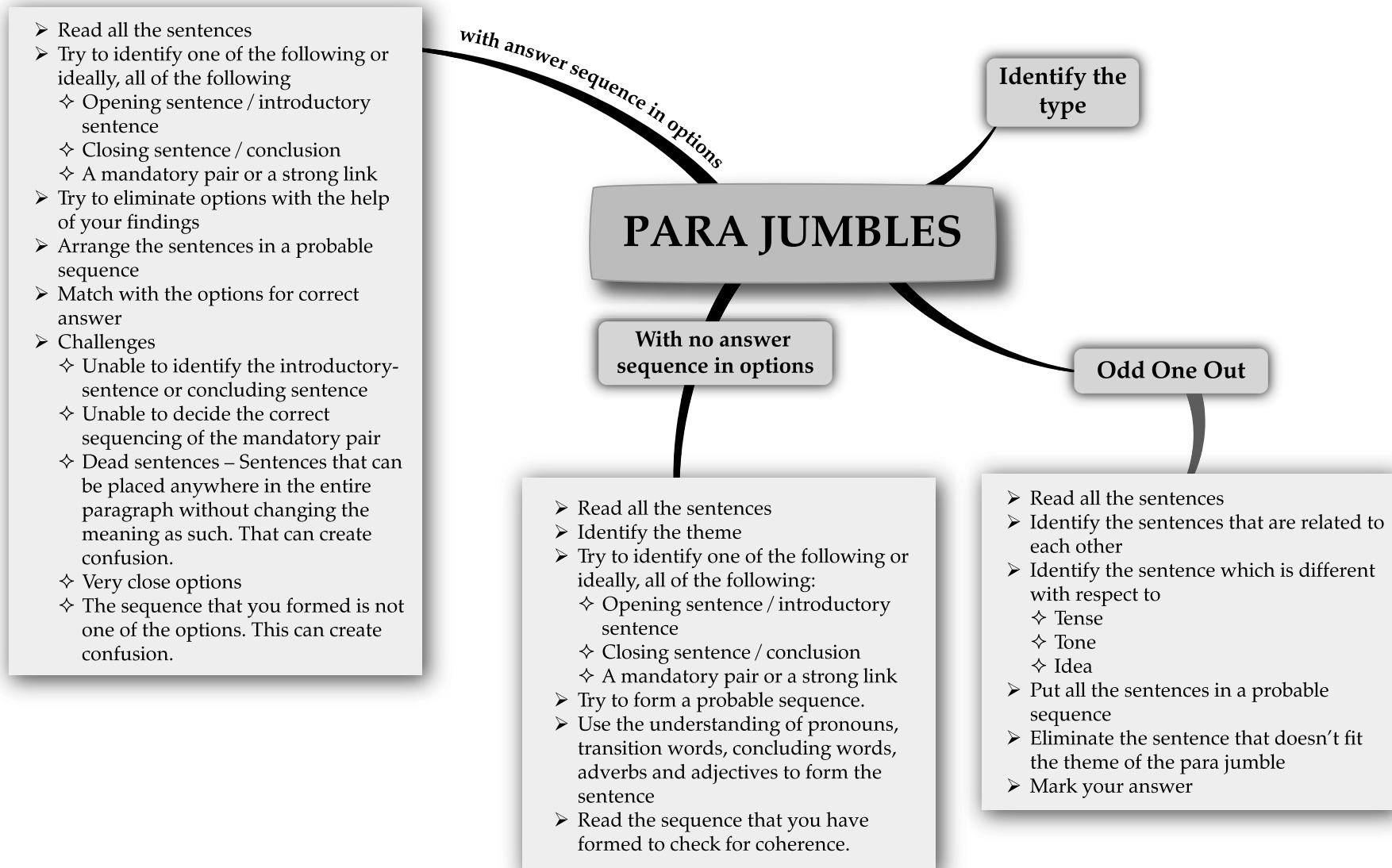
VERBAL APTITUDE MIND MAPS

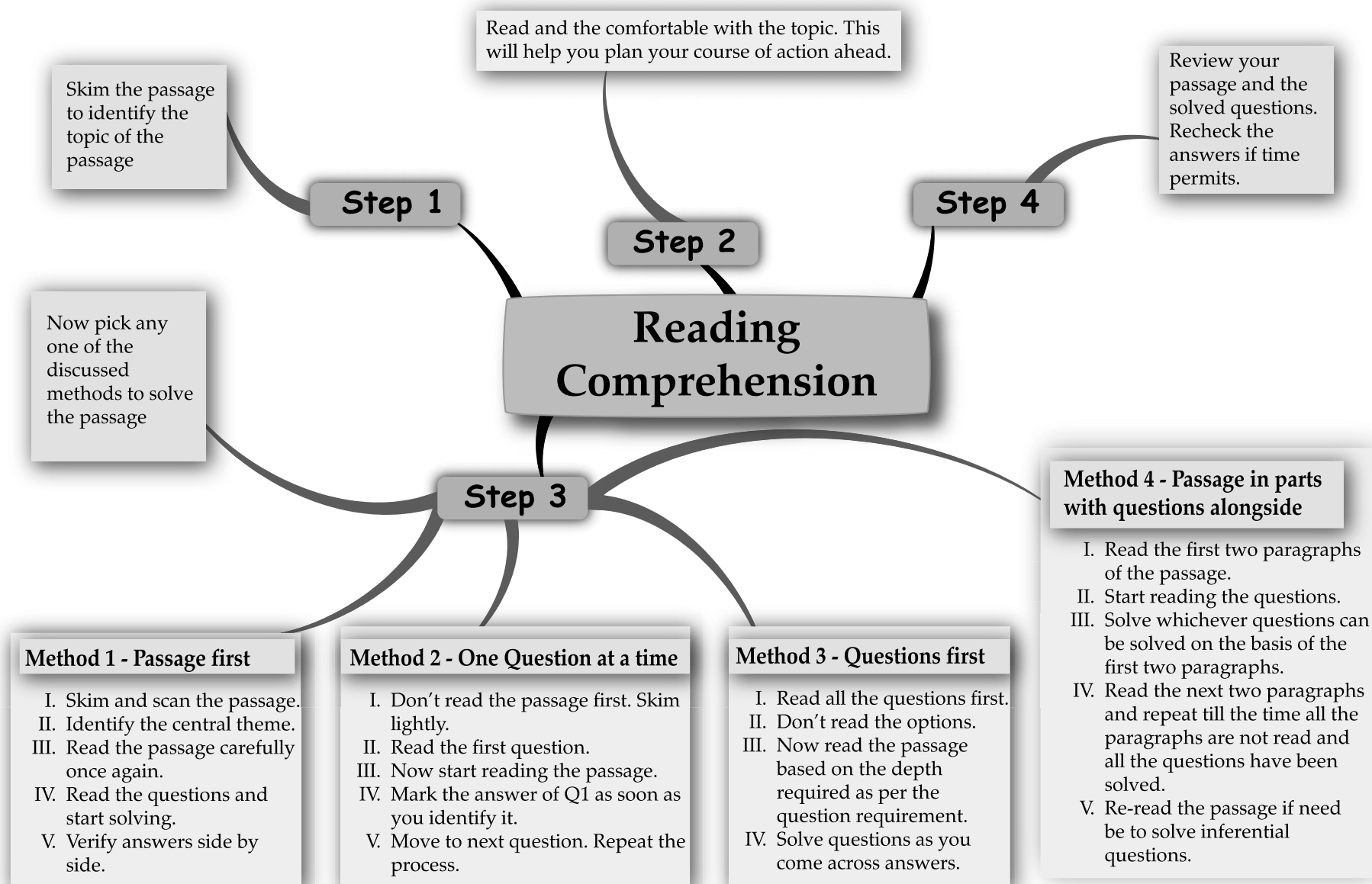




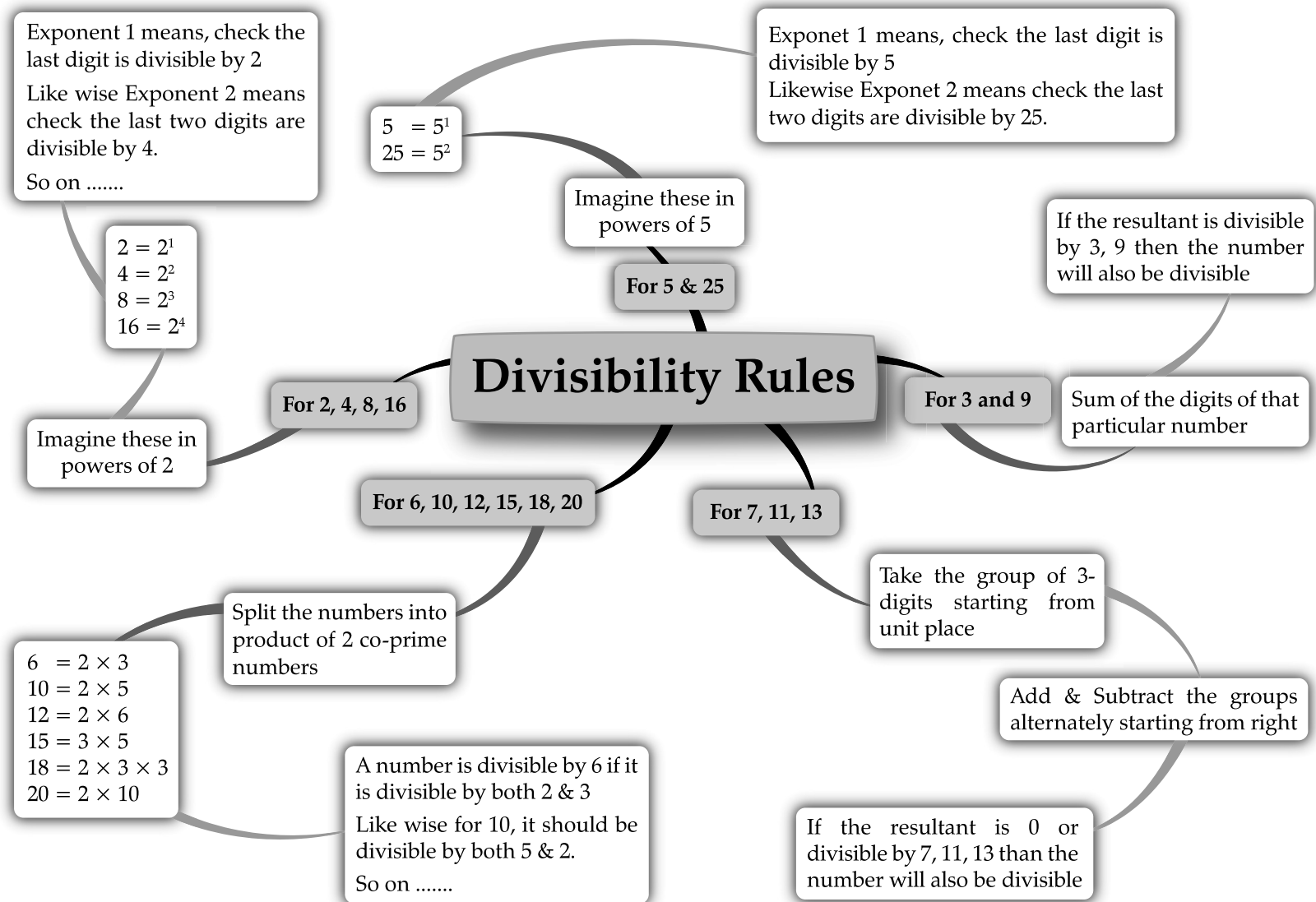


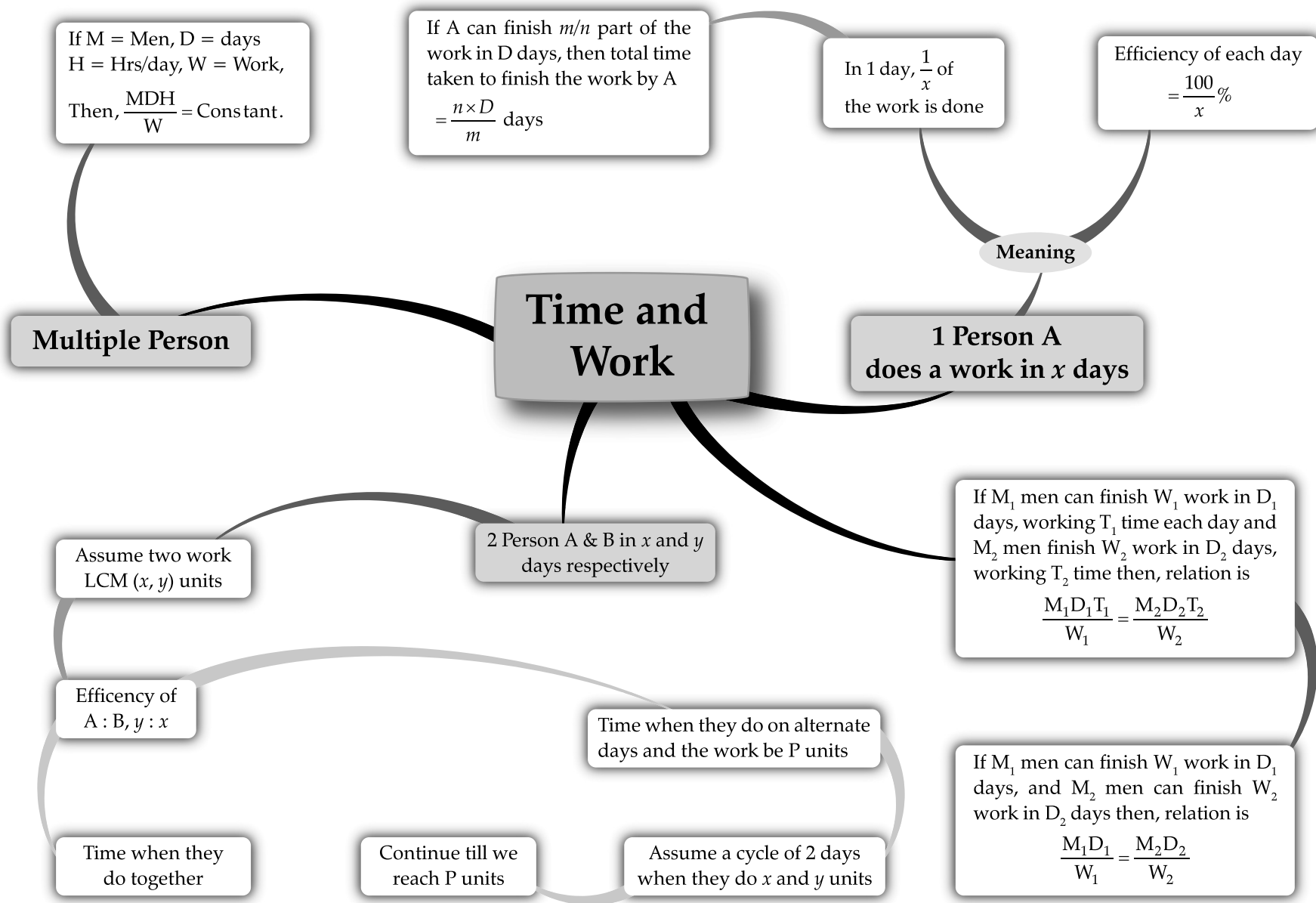


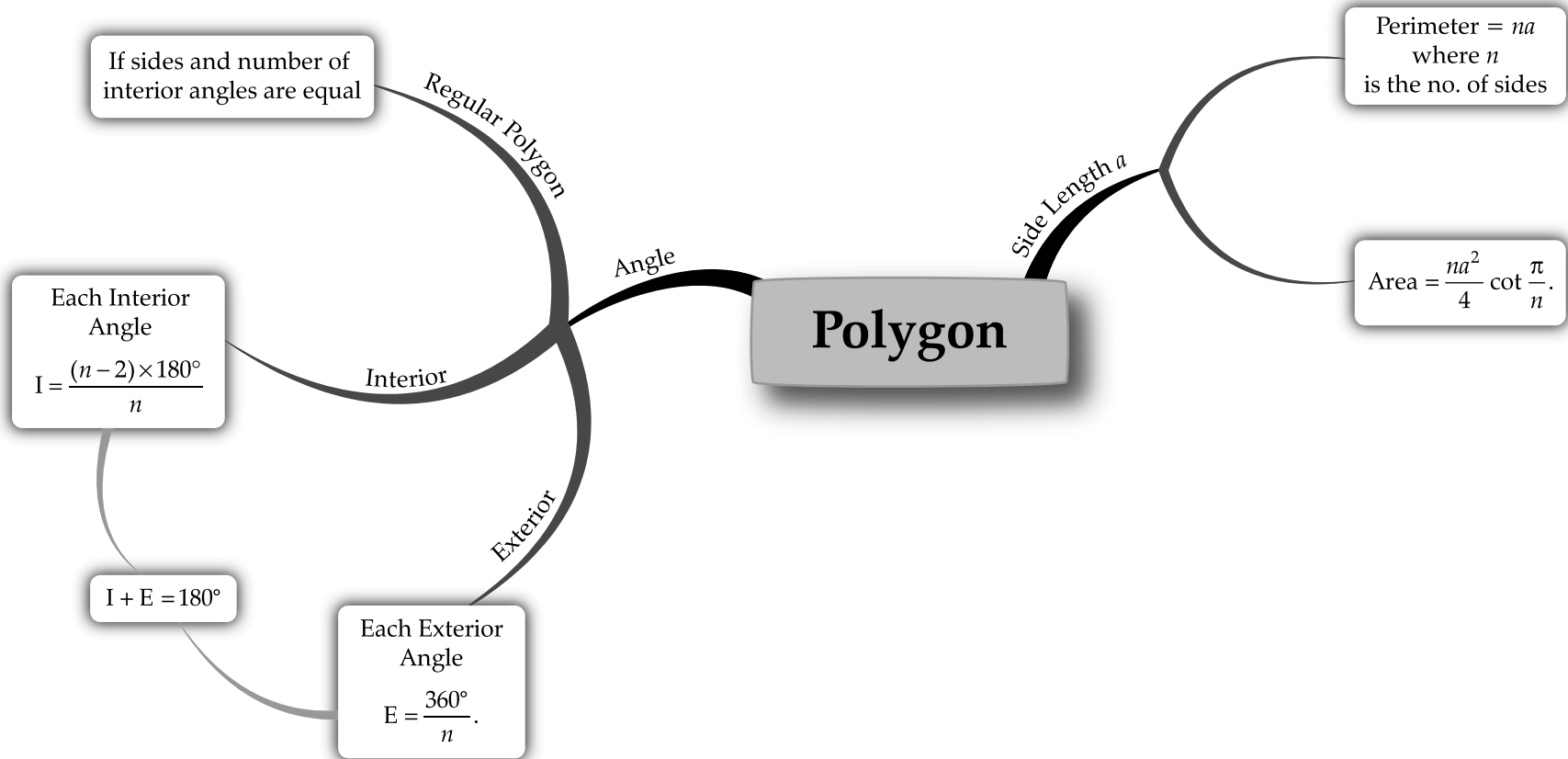


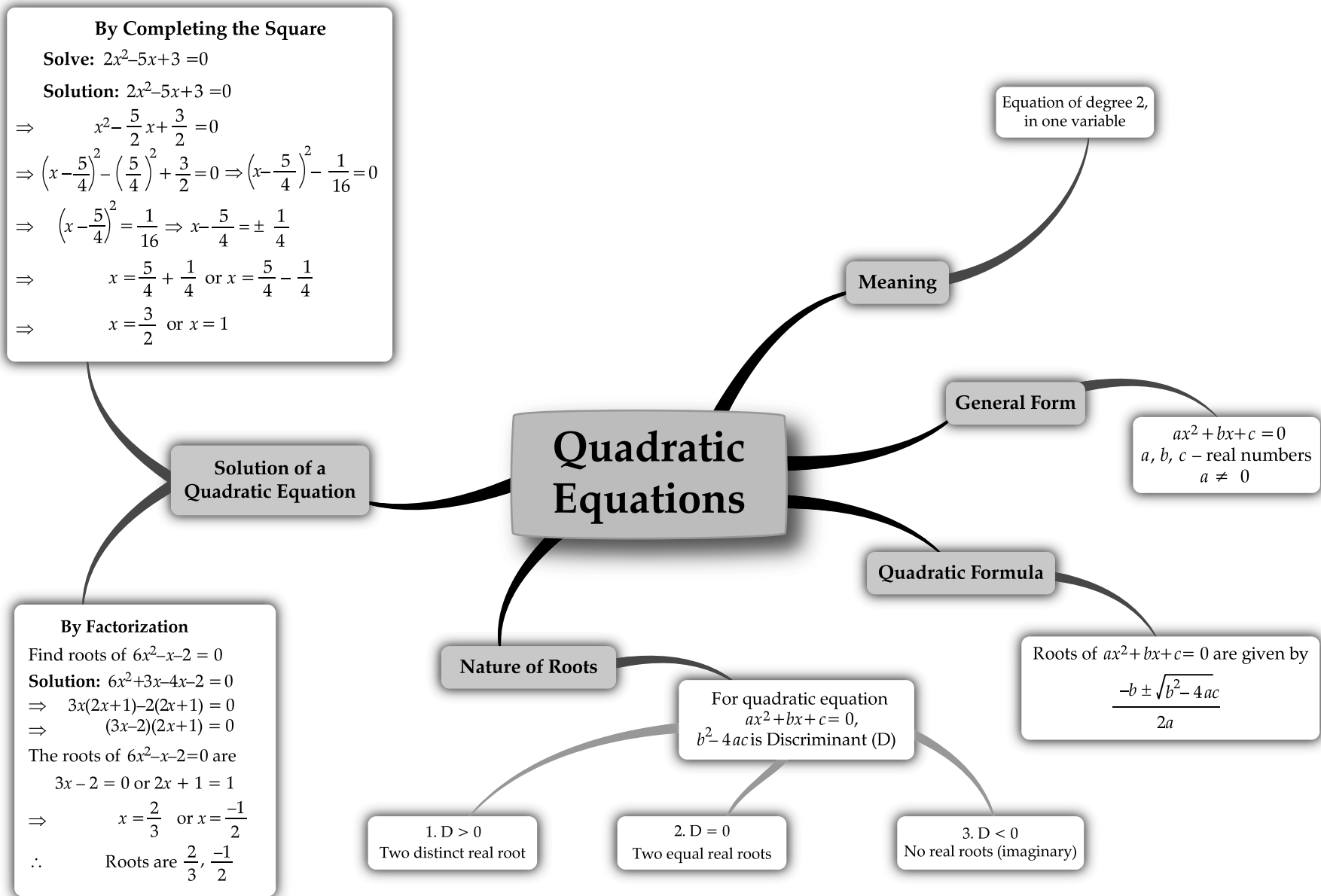


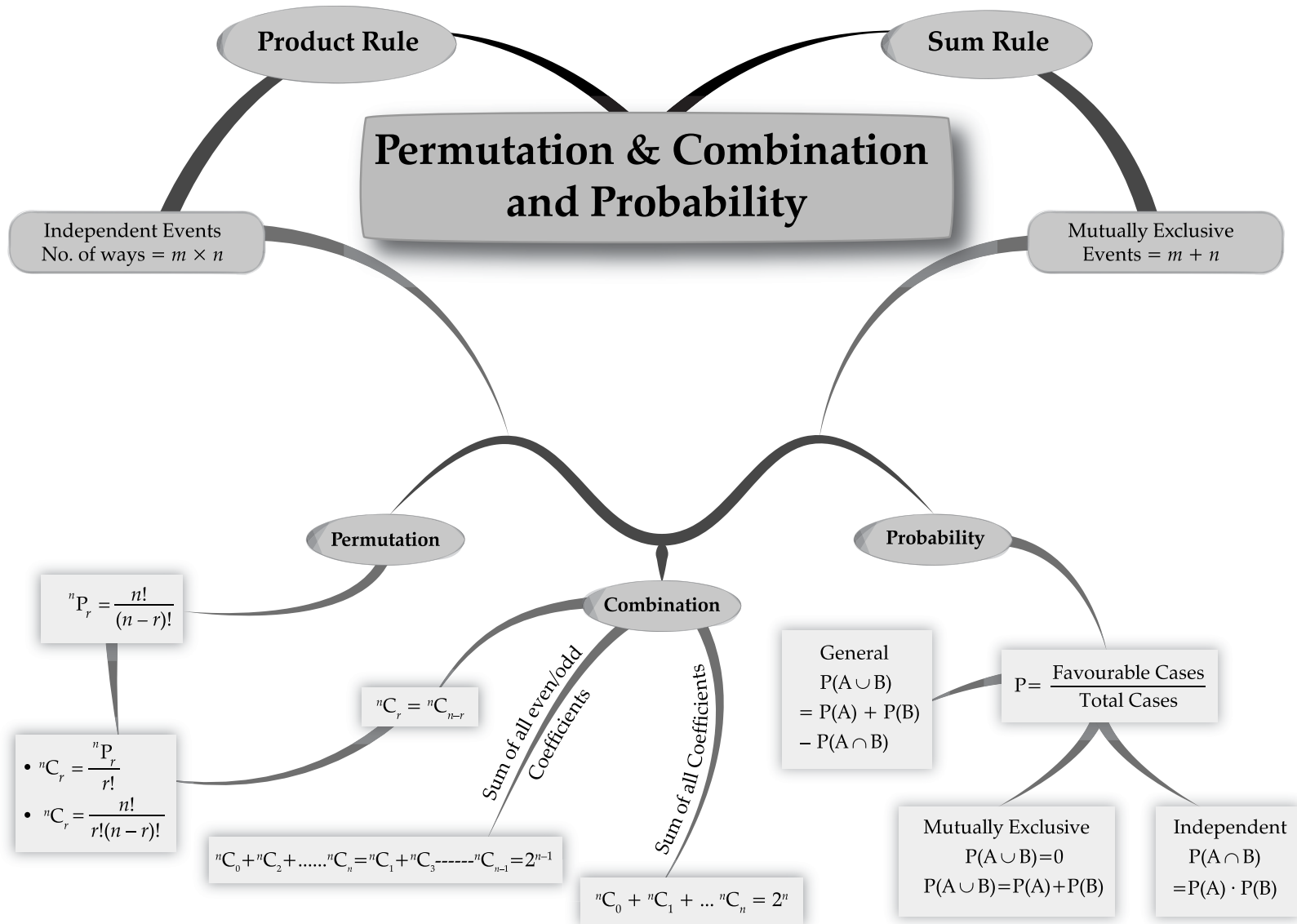
QUANTITATIVE APTITUDE AND ANALYTICAL APTITUDE MIND MAPS

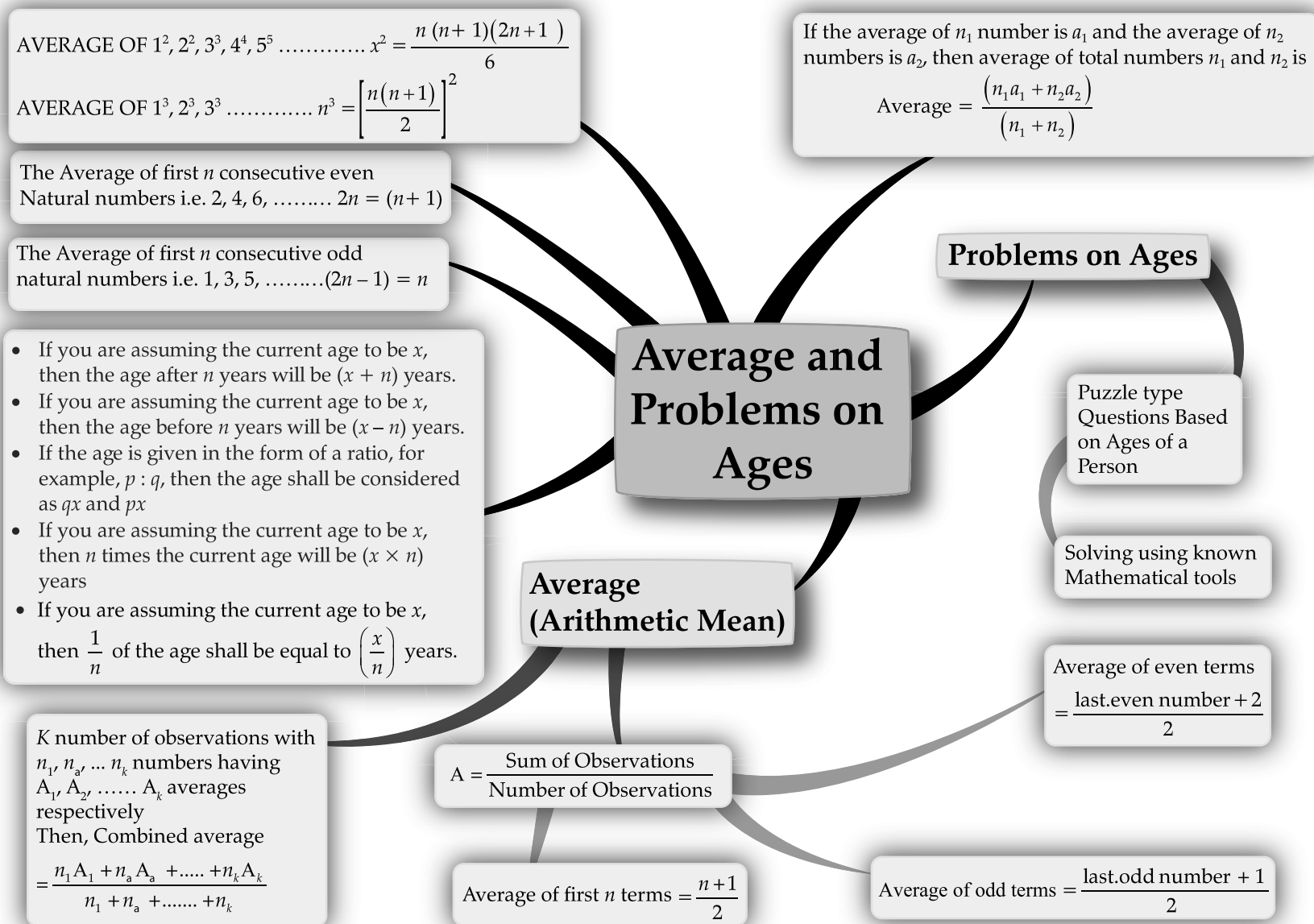


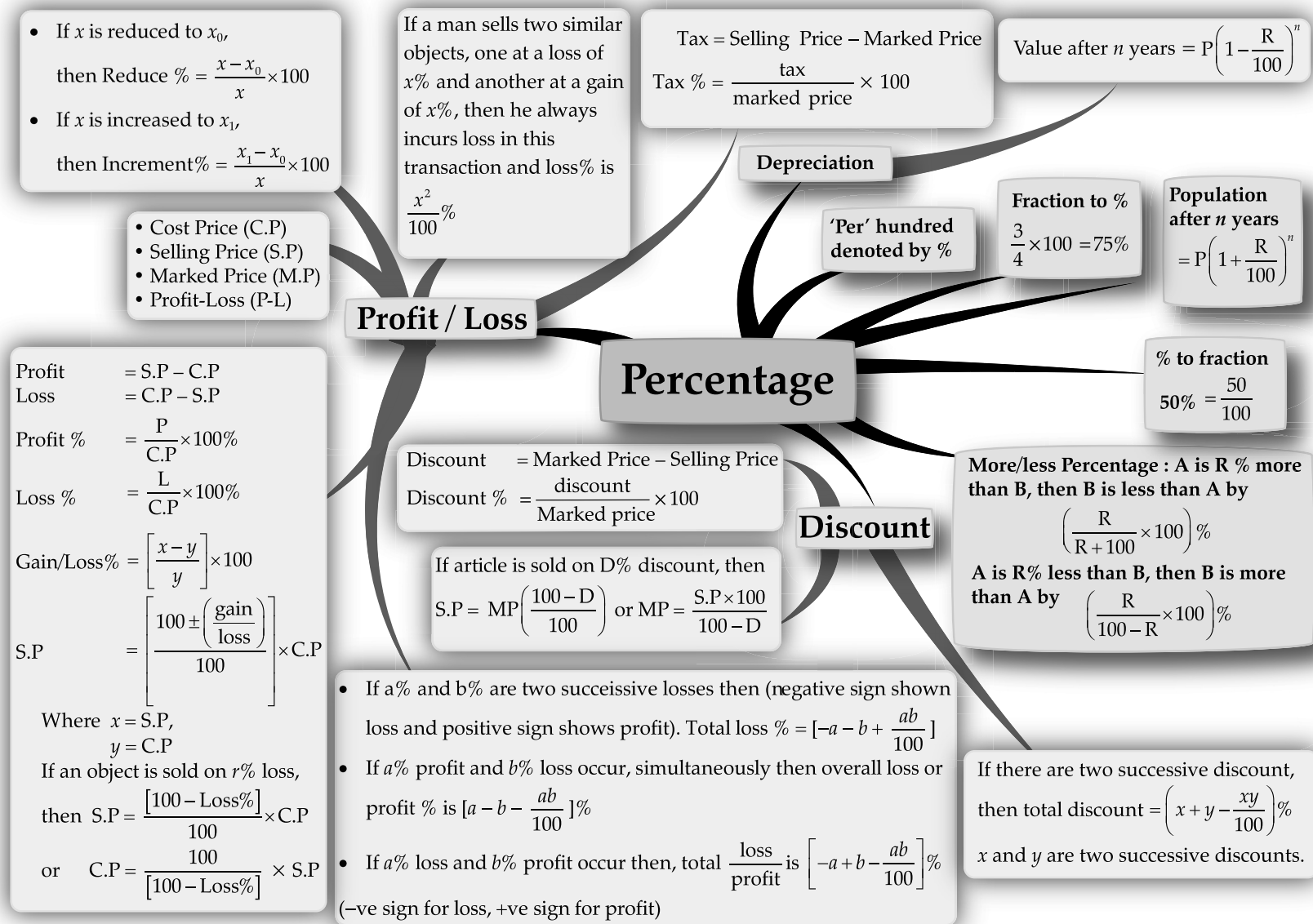


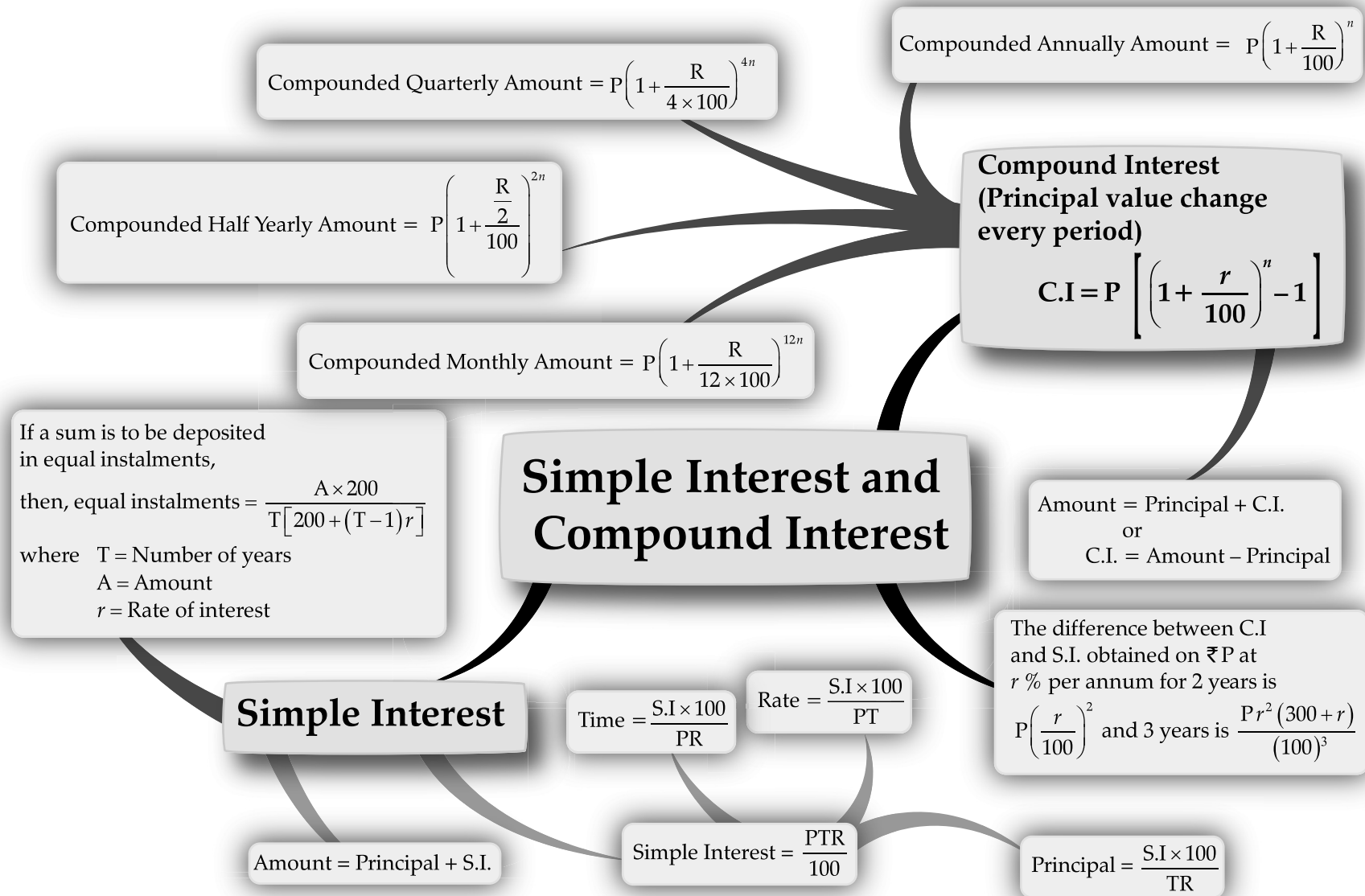


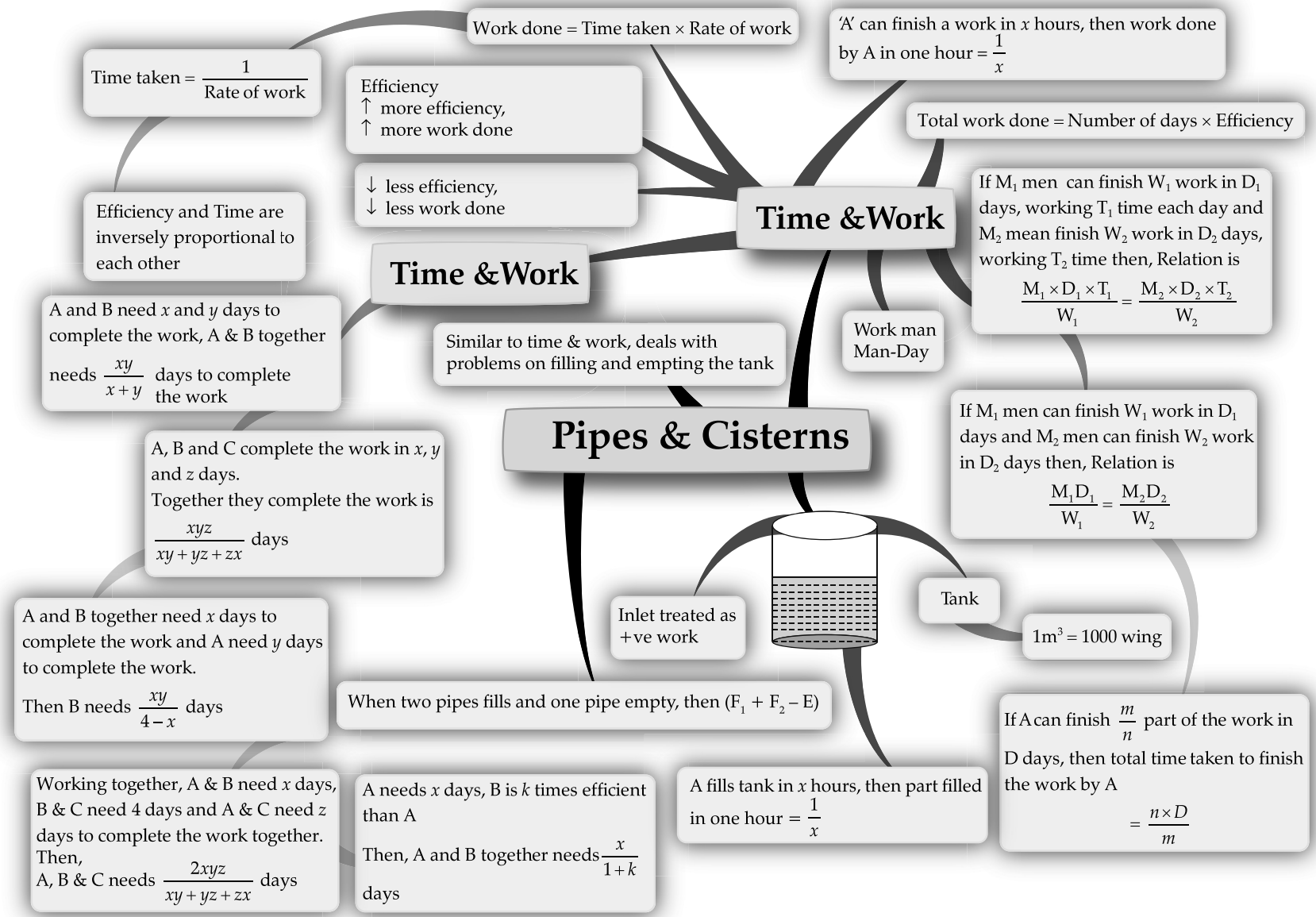












Speed, Time and Distance

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$

$$1 \text{ m/s} = \frac{18}{5} \text{ km/h}$$

Average speed when distance is same

$$A.S = \frac{2xy}{x+y}$$

(where x and y are two speeds)

Average speed when time is same

$$= \frac{(x+y)}{2}$$

(where x and y are two speeds)

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

If $a : b$ is ratio of speed
Then $b : a$ is ratio of time
for the same distance

$$\text{Speed of Boat in still water} = \frac{1}{2}$$

(Downstream speed + upstream speed)

Boat and Stream

Upstream

Downstream

Speed of Boat in still water $\rightarrow x$ km/h
Speed of stream $\rightarrow y$ km/h

$$\text{Speed} = S_b - S_s$$

$$\text{Speed} = S_b + S_s$$

$$\text{Upstream speed} = (x - y) \text{ km/h}$$

$$\text{Downstream speed} = (x + y) \text{ km/h}$$

$$\text{Speed of stream} = \frac{1}{2} (\text{Downstream speed} - \text{Upstream speed})$$

Problem on Trains

Two trains having length a and b with speeds x and y respectively
Then the trains cross each other at

$$\text{Time} = \begin{cases} \text{Opposite} & \left\{ \frac{a+b}{x+y} \right\}, \text{ travelling opposite in direction} \\ \text{Same} & \left\{ \frac{a+b}{x-y} \right\}, \text{ travelling same in direction} \end{cases}$$

$$\text{Total Distance} = \text{Distance} + \text{Length of the train}$$

$$\text{Length of a train} = \text{Speed} \times \text{Time}$$

(when passing by a pole/point)

Distance travelled to cross
Bridge/platform of length y m,
length of the train x m is $(x + y)$

Speed of two trains
Train 1 $- x$ km/h
Train 2 $- y$ km/h
• Same direction
Relative speed $= (x - y)$ km/h

• Opposite direction
Relative speed $= (x + y)$ km/h

Distance traveled to cross
Pole/man = length of the
train it self

$$\text{Average speed} = \frac{\text{Downstream speed} \times \text{Upstream speed}}{\text{Speed in still water}}$$

